Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	4 march 2026
Team ID	SWTID-2026-2725
Project Name	SpendWise – Simple CRUD & JWT Backend
Maximum Marks	4 Marks

❖Technical Architecture:

The system follows a **three-layer architecture** consisting of the presentation layer, application layer, and data layer. This structure ensures better performance, security, and maintainability of the application.

❖Table 1 – Components & Technologies:

S.No	Component	Description	Technology
1	Presentation Layer	Provides user interface for interacting with the system and viewing expense records	React.js / HTML / CSS
2	Business Logic Layer	Processes user requests and performs expense calculations and validations	Node.js, Express.js

S.No	Component	Description	Technology
3	Data Storage	Stores user information, income records, and expense transactions	MongoDB
4	Authentication Module	Manages login, registration, and password encryption	JWT, bcrypt
5	Configuration Module	Handles environment variables and project configuration	dotenv
6	Service Layer	Handles communication between client requests and database operations	RESTful APIs

❖ **Table 2 – Application Characteristics**

S.No	Characteristics	Description	Technology
1	Modular Design	Application is divided into independent modules for easy maintenance	Node.js Project Structure
2	Secure Access	Protects user data through token based authentication	JWT
3	Database Connectivity	Efficient communication	Mongoose

S.No	Characteristics	Description	Technology
		between backend server and database	
4	RESTful Communication	Data is exchanged using REST architecture	HTTP Methods (GET, POST, PUT, DELETE)
5	Scalable Architecture	System can be expanded with new modules and features easily	Express.js
6	Environment Management	Sensitive configuration details stored separately from code	.env File